



README FILE

TITLE OF THE PROJECT - House Price Predictor

OVERVIEW OF THE PROJECT - It is a system which tells about the price of a house based on the space, number of bedrooms and timespan of the house. It automatically predict the price of the house with the help of simple calculation and a formula.

FEATURES –

1. It allows to predict the price of the house with the help of libraries and formulas and on the basis of area of the house, number of bedrooms a house contains and timespan of the house.
2. Allows user to input the details about the type of house and customise the area, bedrooms and timespan.
3. Gives separate values of the variable such as 'w1', 'w2', 'w3', 'b' which are used in the formula.
4. Calculates the rate of the house in seconds once the input is given.
5. Each and every step is clearly mentioned making the code easy to understand.
6. Comments are added for the better understanding.

TECHONOLOGIES AND TOOLS USED –

1. 'pandas' and 'LinearRegression' are the libraries which are imported.
2. Dictionary is used to store the details about the house such as area, number of bedrooms, and the timespan of the house.
3. Multiple values are stored in the List.
4. Data is converted into tabular form using 'pd.DataFrame()'.
`pd.DataFrame()`
5. Training function is used.
6. Input function is used to enter values and customise the house according to the user.
7. Rate predictor 'model.predict()' is used to predict the price of house.

STEPS TO INSTALL AND RUN THE PROJECT –

1. Install python and VScode so that you can run the python code.
2. Firstly, run the code and then enter the details of the house.
3. Then the rate predictor will calculate and predict the price of the house.
4. Finally, the rate will be calculated and will be visible on the screen.

INSTRUCTIONS FOR TESTING –

1. Enter all the details the code ask for.
2. Donot skip any step otherwise it will show error.
3. Read the code carefully and follow each step line wise.

```
[10]: # IMPORT LIBRARIES
import pandas as pd
from sklearn.linear_model import LinearRegression

# ENTERING DATA
details = {
    'space': [100, 160, 200, 250, 350, 800, 1000],
    'bedrooms': [1, 2, 2, 3, 4, 4, 5],
    'timespan': [10, 8, 7, 6, 5, 4, 2],
    'rate': [7, 12, 16, 22, 30, 45, 60]
}

# DATA IN TABULAR FORM
df = pd.DataFrame(details)

# INPUT (X) AND OUTPUT (Y)
X = df[['space', 'bedrooms', 'timespan']]
Y = df['rate']
```

```
# MODEL CREATION
# FORMULA USED : rate = w1.space + w2.bedrooms + w3.timespan + b
model = LinearRegression()

# TRAINING FUNCTION
model.fit(X,Y)

# LEARNED VALUES

# VALUE OF w1, w2, w3
print(model.coef_)

# VALUE OF b
print(model.intercept_)

# INPUTS FROM USERS
space = int(input("Space of the house:"))
bedrooms = int(input("Number of bedrooms needed:"))
timespan = int(input("Enter the time span of house:"))

# RATE PREDICTION
a = model.predict([[space,bedrooms,timespan]])
```

```
# FINAL RATE
print("Price of the house:",round(a[0],4),"lakhs")
```

```
[ 0.03513258  3.40246263 -0.96259379]
8.642577391185558
Space of the house: 100
Number of bedrooms needed: 7
Enter the time span of house: 9
Price of the house: 27.3097 lakhs
```